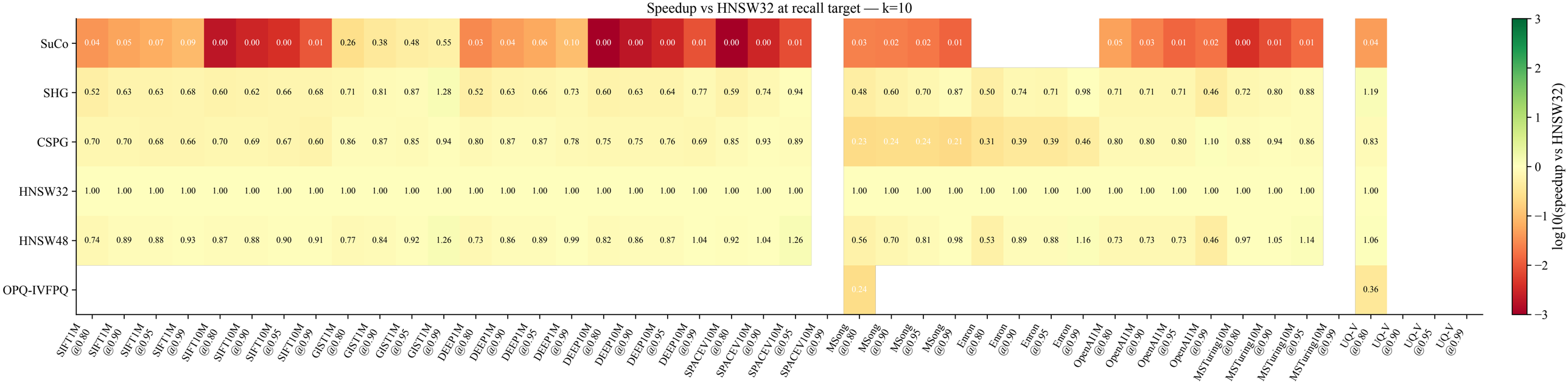
Speedup vs HNSW32 at recall target — k=10
SuCo
SHG
CSPG
HNSW32
HNSW48
OPQ-IVFPQ
0.04 0.05 0.07 0.09 0.00 0.00 0.00 0.01 0.26 0.38 0.48 0.55 0.03 0.04 0.06 0.10 0.00 0.00 0.00 0.01 0.00 0.00 0.01 0.03 0.02 0.02 0.01 0.05 0.03 0.01 0.02 0.00 0.01 0.01 0.04
0.52 0.63 0.63 0.68 0.60 0.62 0.66 0.68 0.71 0.81 0.87 1.28 0.52 0.63 0.66 0.73 0.60 0.63 0.64 0.77 0.59 0.74 0.94 0.48 0.60 0.70 0.87 0.50 0.74 0.71 0.98 0.71 0.71 0.71 0.46 0.72 0.80 0.88 1.19
0.70 0.70 0.68 0.66 0.70 0.69 0.67 0.60 0.86 0.87 0.85 0.94 0.80 0.87 0.87 0.78 0.75 0.75 0.76 0.69 0.85 0.93 0.89 0.23 0.24 0.24 0.21 0.31 0.39 0.39 0.46 0.80 0.80 0.80 1.10 0.88 0.94 0.86 0.83
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.74 0.89 0.88 0.93 0.87 0.88 0.90 0.91 0.77 0.84 0.92 1.26 0.73 0.86 0.89 0.99 0.82 0.86 0.87 1.04 0.92 1.04 1.26 0.56 0.70 0.81 0.98 0.53 0.89 0.88 1.16 0.73 0.73 0.73 0.46 0.97 1.05 1.14 1.06
0.24 0.36
SIFT1M @0.80
SIFT1M @0.90
SIFT1M @0.95
SIFT1M @0.99
SIFT10M @0.80
SIFT10M @0.90
SIFT10M @0.95
SIFT10M @0.99
GIST1M @0.80
GIST1M @0.90
GIST1M @0.95
GIST1M @0.99
DEEP1M @0.80
DEEP1M @0.90
DEEP1M @0.95
DEEP1M @0.99
DEEP10M @0.80
DEEP10M @0.90
DEEP10M @0.95
DEEP10M @0.99
SPACEV10M @0.80
SPACEV10M @0.90
SPACEV10M @0.95
SPACEV10M @0.99
MSong @0.80
MSong @0.90
MSong @0.95
MSong @0.99
Enron @0.80
Enron @0.90
Enron @0.95
Enron @0.99
OpenAI1M @0.80
OpenAI1M @0.90
OpenAI1M @0.95
OpenAI1M @0.99
MSTuring10M @0.80
MSTuring10M @0.90
MSTuring10M @0.95
MSTuring10M @0.99
UQ-V @0.80
UQ-V @0.90
UQ-V @0.95
UQ-V @0.99
log10(speedup vs HNSW32)
3
2
1
0
−1
−2
−3